Milestone 2 - STEAMingHot

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Project Goals

Our visualization is intended to help game developers understand what makes a game become popular based purely on data and statistics. To achieve this, we will rely on different visualization methods to evaluate different aspects of success. The three core aspects that are part of our **minimum viable product (MVP)** are the following:

- A way to inspect the data for a specific game, and show how that game compares to others in the market. We already created a basic template for that section in our demo website (Fig 1).
- A global overview of the steam market, with a visualization showing statistics on all (relevant) games, and the option for the user to modify the statistics they want to see. For that we could use distribution graphs, violin plots, and hexbin. The latter was used in the demo (Fig 2), but will probably be replaced as it fits poorly with the skewed nature of the dataset.
- A race chart to show how the market evolves. For now, our dataset does not provide enough data to see the evolution of prices, users or scores, but we can implement a race chart that shows the top categories being released each year, which effectively shows the way the competition evolves.

With enough time we plan to expand the MVP with **extra features**:

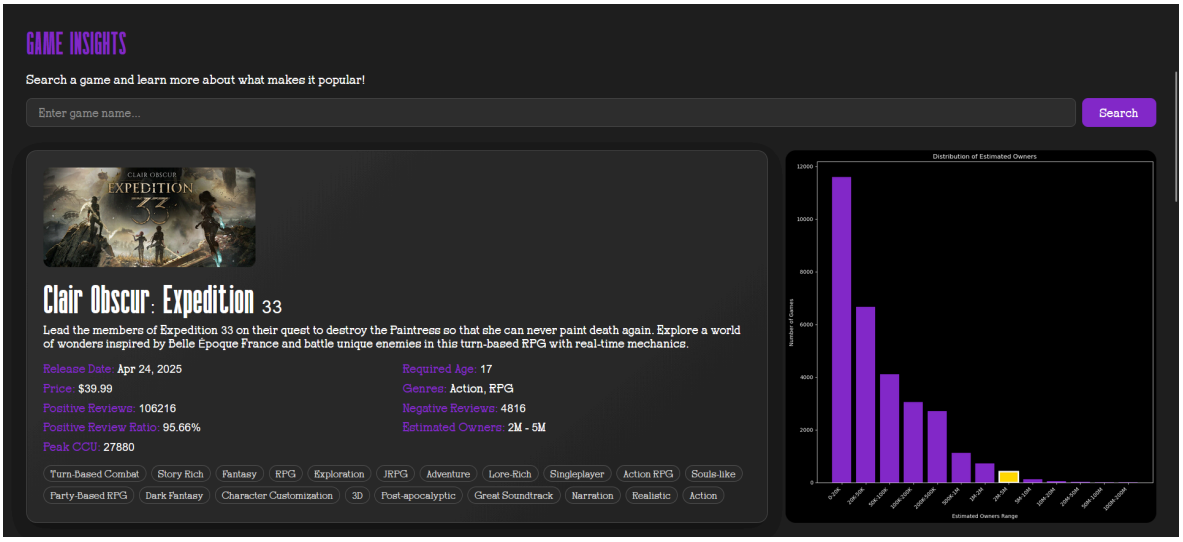
- A map which could show regional tendencies around the world, to help select the target audience for a new game. This will require extending the dataset with regional data.
- A brushable scatterplot matrix (and also barplots), which could show a lot of insights at the same time.
- A graph showing the correlation between the price reduction and the percentage of sales peak. This will also require a dataset augmentation.

Next steps

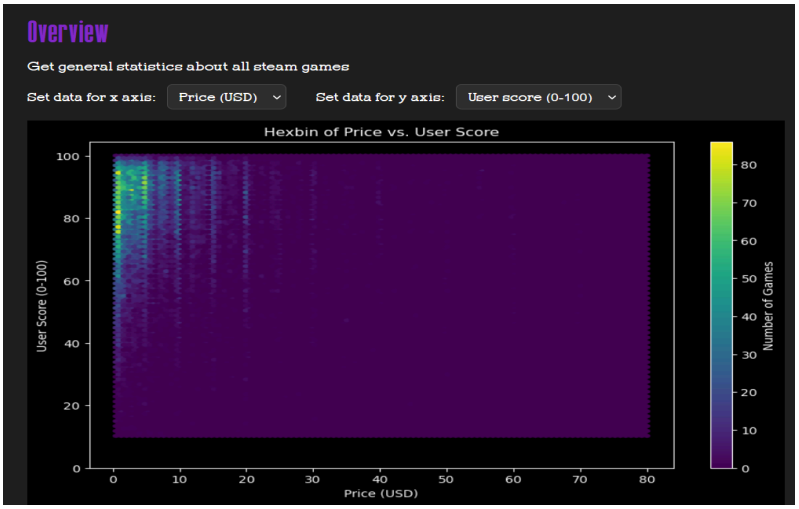
The template for the website is already ready. The next steps toward the MVP are the following:

1. Create a preprocessing pipeline which removes some irrelevant obscure games.
2. Implement the three visualizations planned for the MVP. As we are a group of three, the distribution will be fairly easy.
3. Add more interaction to the base visualization (e.g give options to customize the graphs with different axis values).
4. If time remains, implement extra features.

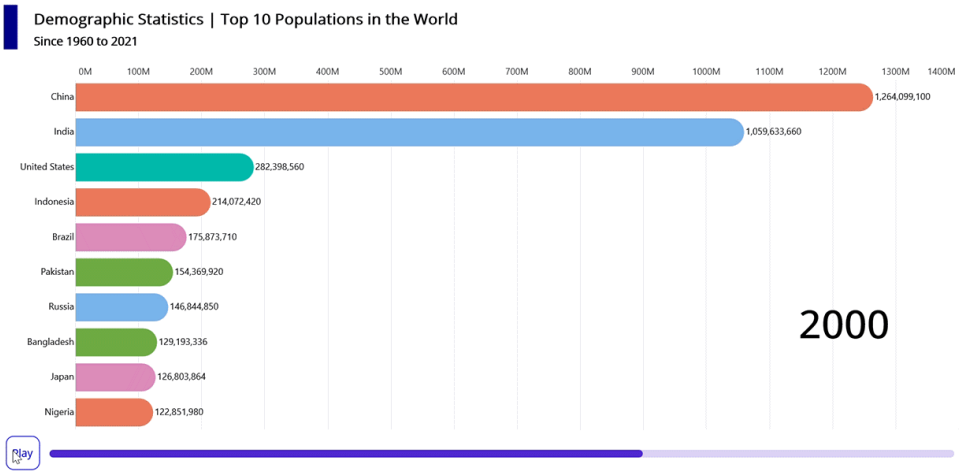
Sketches



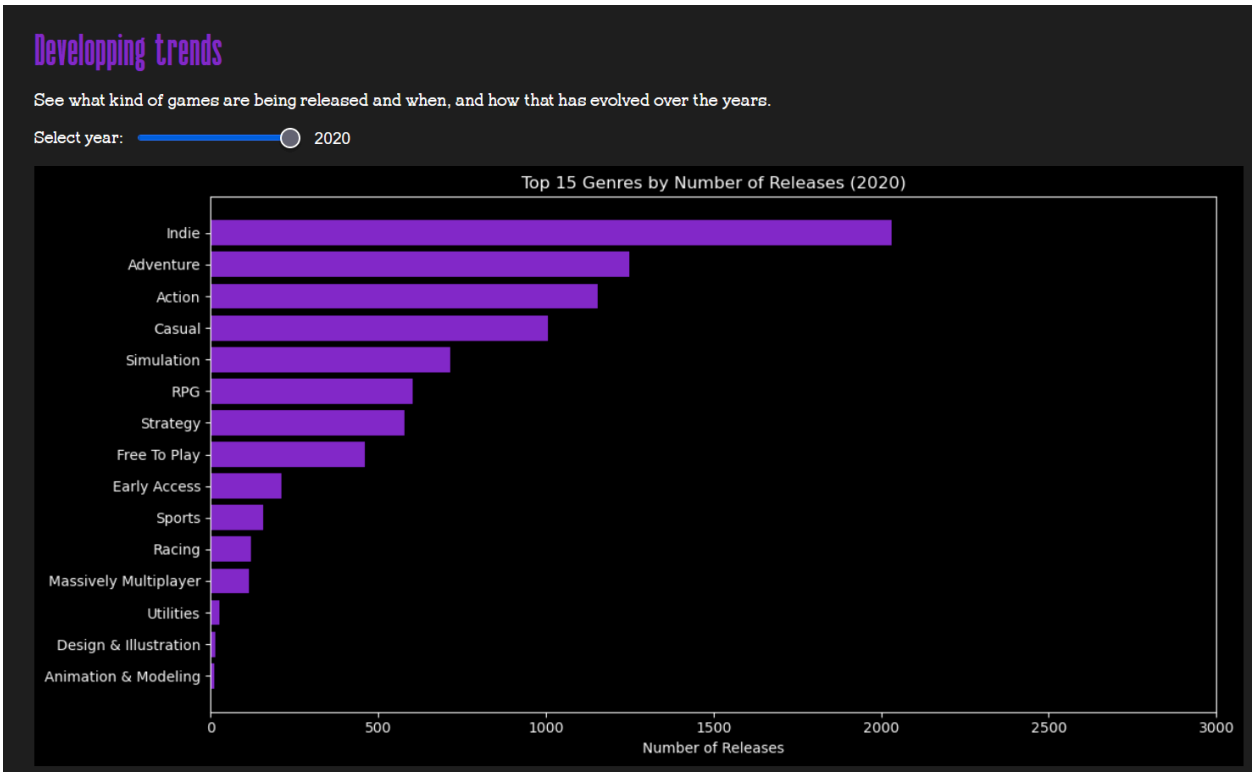
(Fig 1) Actual design of the "Game Insights" section of our demo website



(Fig. 2) An overview of the steam dataset, comparing the relation between price and score



(Fig.3) Expected kind of result with the race chart



(Fig. 4) A placeholder bar plot obtained with our data

Steam Reviews Language Breakdown

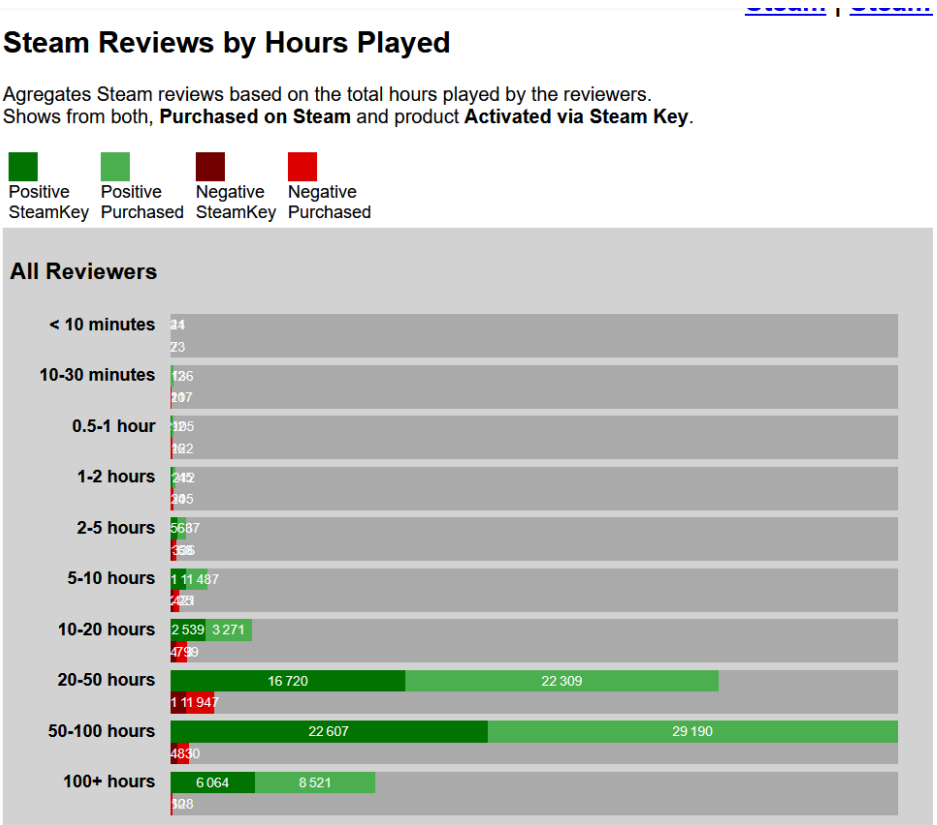
Purchase Type: All

Shows **all reviews** from both **Purchased on Steam** and product **Activated via Steam Key**.
Click on the header to *sort the table*. Click on the language row to *read reviews*.

Languages: ❌ Unsupported | ✅ Supported

Portion ▼	Language	Total Reviews (262 204)	Total Positive (250 347)	Total Negative (11 857)	Review Score 95.48% + (Overwhelmingly Positive)
50.27%	✅ English	131 792	127 487	4 305	97% + (Overwhelmingly Positive)
14.15%	✅ Simplified Chinese	37 093	32 242	4 851	87% + (Very Positive)
8.60%	✅ French	22 527	22 265	262	99% + (Overwhelmingly Positive)
5.93%	✅ Russian	15 546	14 593	953	94% + (Very Positive)
3.49%	✅ Spanish - Spain	9 131	8 988	143	98% + (Overwhelmingly Positive)
3.38%	✅ Portuguese - Brazil	8 847	8 763	84	99% + (Overwhelmingly Positive)
3.06%	✅ German	8 006	7 723	283	96% + (Overwhelmingly Positive)
2.17%	✅ Korean	5 666	5 406	260	95% + (Overwhelmingly Positive)
1.98%	✅ Polish	5 176	5 081	95	98% + (Overwhelmingly Positive)
1.69%	✅ Traditional Chinese	4 413	4 220	193	96% + (Overwhelmingly Positive)
1.36%	✅ Italian	3 549	3 492	57	98% + (Overwhelmingly Positive)
0.94%	✅ Spanish - Latin America	2 439	2 415	24	99% + (Overwhelmingly Positive)

(Fig. 5) A plot from SteamScout that could be used as inspiration



(Fig. 6) Another plot from SteamScout that could be used as inspiration

Tools needed

- D3.js is a must for web data visualization.
- Steam game scrapper is the script we will use to acquire our data.
- There exist numerous tools and applications that can provide or help generate racing charts. Here are some examples : https://www.youtube.com/watch?v=j20Kfxbzrqs https://www.zoho.com/sheet/race-charts.html https://flourish.studio/visualisations/bar-chart-race/
    - We will take inspiration from them to code our own racing chart.
- https://togeproductions.com/SteamScout/ provides all the information needed for this graph. We would also possibly need the Data lecture, Tabular Data lecture. SteamScout takes information from many sources, such as Steam250 (https://club.steam250.com/app/1903340) that gathers and categorizes the reviews.
- SteamDB will be our main inspiration for this visualization as it does it perfectly already. https://steamdb.info/app/2868840/charts/
- https://www.codecks.io/blog/2021/best-steam-tools-for-game-developers/